

Lab 02

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Task 1a

```
x = 15.5 + 3.175i
```

```
x =  
15.5000 + 3.1750i
```

```
r1=real(x) %real value
```

```
r1 =  
15.5000
```

```
i1 = imag(x) %imaginary value
```

```
i1 =  
3.1750
```

```
c1=conj(x) %complex conjugate
```

```
c1 =  
15.5000 - 3.1750i
```

```
a1 = abs(x) %magnitude
```

```
a1 =  
15.8218
```

```
p1 = angle(x) %phase angle
```

```
p1 =  
0.2020
```

Task 1b

```
[r, im]=pol2cart(45*pi/100, 5)
```

```
r =  
0.7822  
im =  
4.9384
```

```
x1=r +im*j
```

```
x1 =  
0.7822 + 4.9384i
```

```
r2=real(x1) %real value
```

```
r2 =
```

0.7822

```
i2 = imag(x1) %imaginary value
```

i2 =
4.9384

```
c2=conj(x1) %complex conjugate
```

c2 =
0.7822 - 4.9384i

```
a2 = abs(x1) %magnitude
```

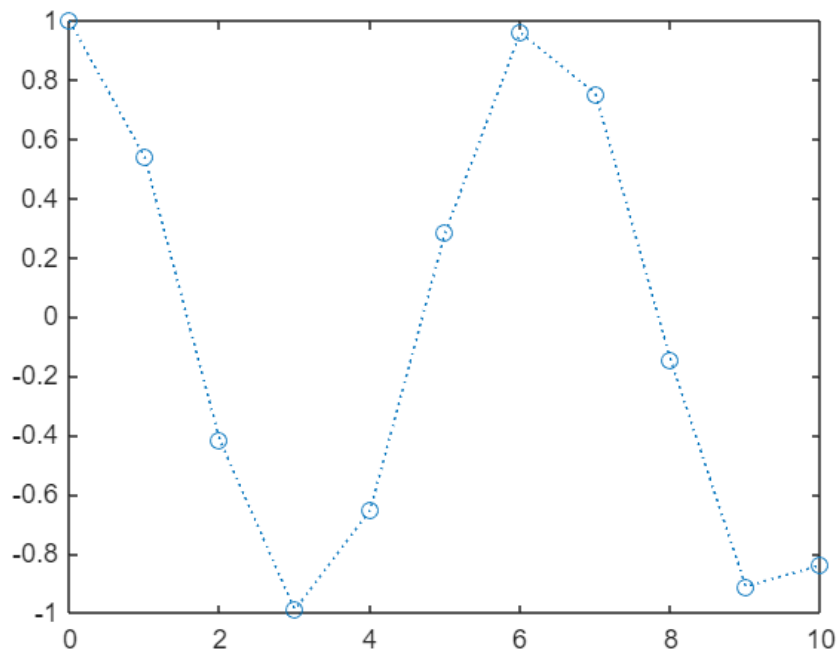
a2 =
5.0000

```
p2 = angle(x1) %phase angle
```

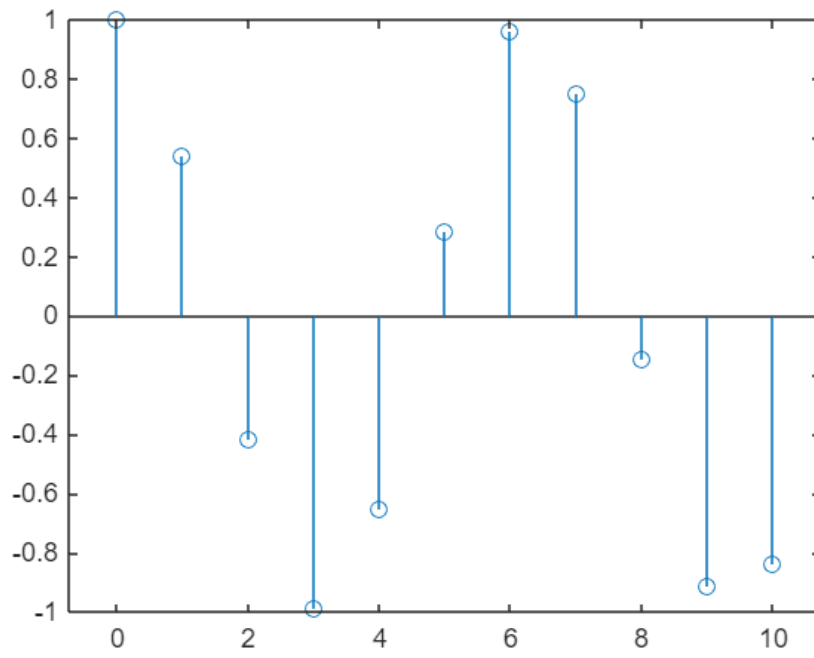
p2 =
1.4137

Try 1

```
n = 0:10;  
y = cos(n);  
figure; plot(n, y, 'o');
```



```
figure; stem(n, y);
```

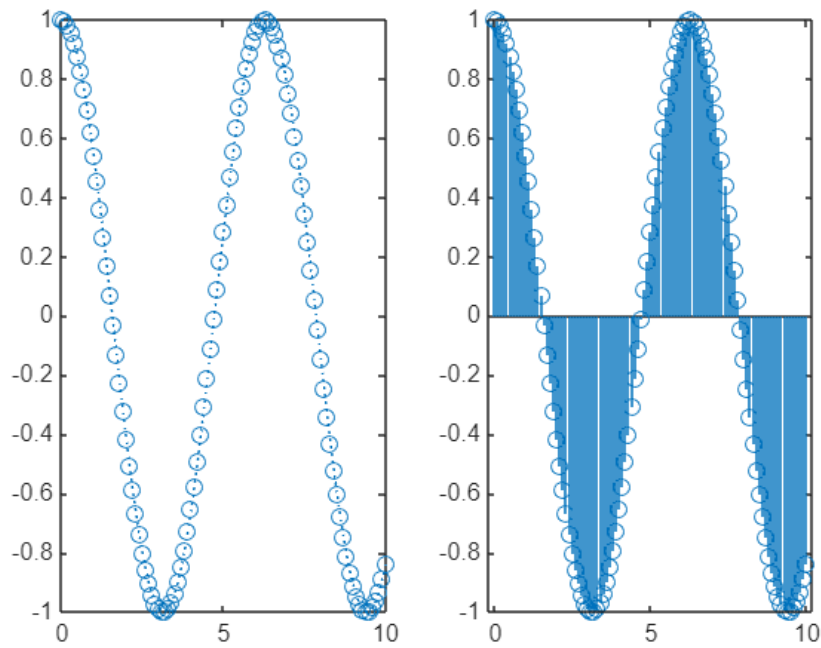


Try 2

```

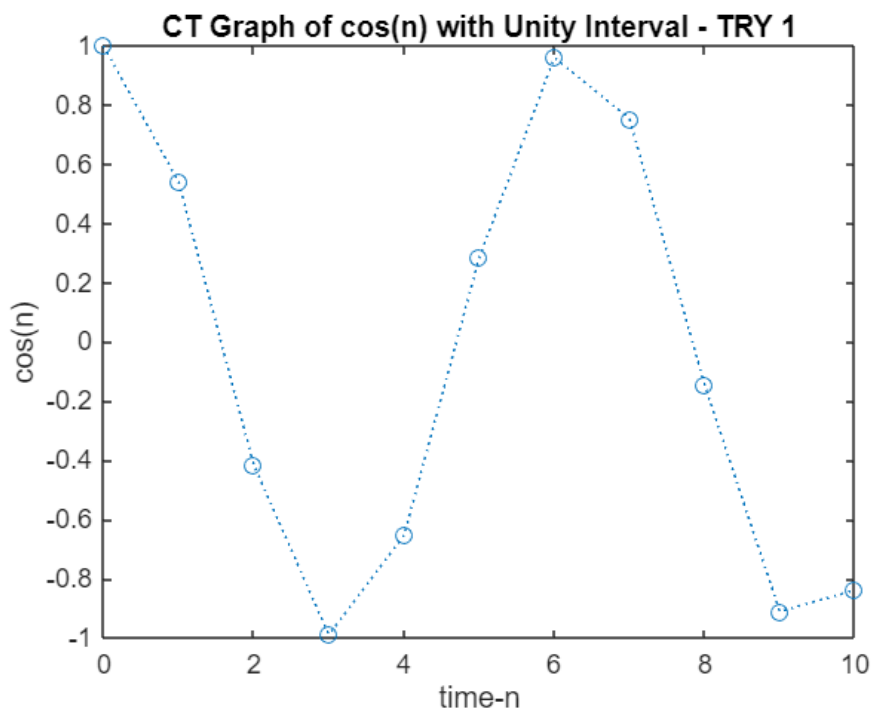
Ts = 0.1;    % Defining sampling time, the time interval between two consecutive
samples
Fs = 1/Ts;   % Defining sampling frequency, the number of samples per second
t = 0:Ts:10; % Defining time span
y = cos(t);  % Obtaining a discrete-time signal
figure;      % Using subplots to depict two plots in one figure
subplot 121; plot(t, y, ':o');
subplot 122; stem(t, y);

```



Task 2a

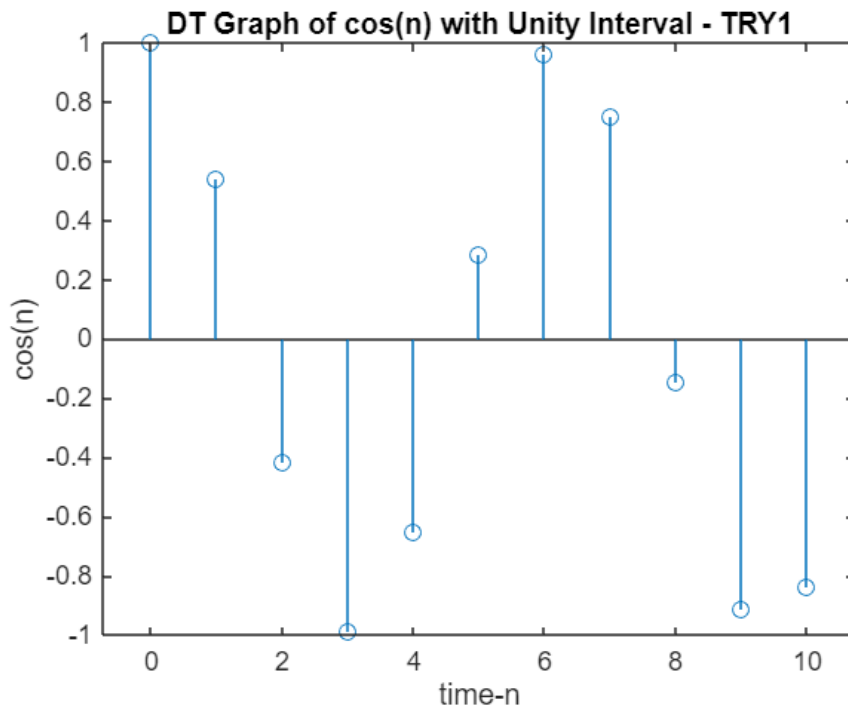
```
n = 0:10;
y = cos(n);
figure; plot(n, y, ':o');title('CT Graph of cos(n) with Unity Interval - TRY 1')
xlabel('time-n')
ylabel('cos(n)')
```



```

stem(n, y)
title('DT Graph of cos(n) with Unity Interval - TRY1')
xlabel('time-n')
ylabel('cos(n)')

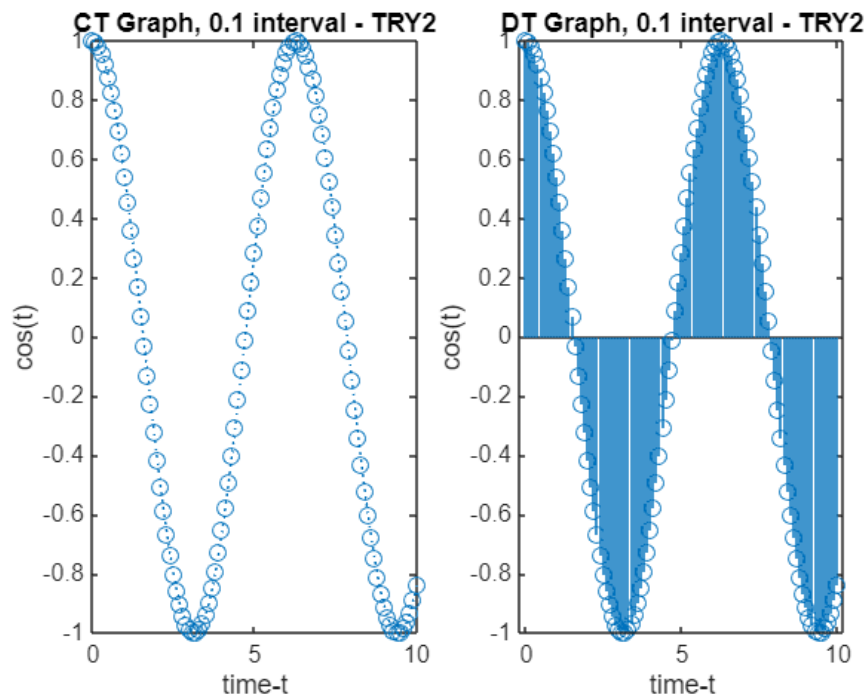
```



```

Ts = 0.1;
Fs = 1/Ts;
t = 0:Ts:10;
y = cos(t);
subplot 121;
plot(t, y, 'o')
title('CT Graph, 0.1 interval - TRY2')
xlabel('time-t')
ylabel('cos(t)')
subplot 122;
stem(t,y)
title('DT Graph, 0.1 interval - TRY2')
xlabel('time-t')
ylabel('cos(t)')

```

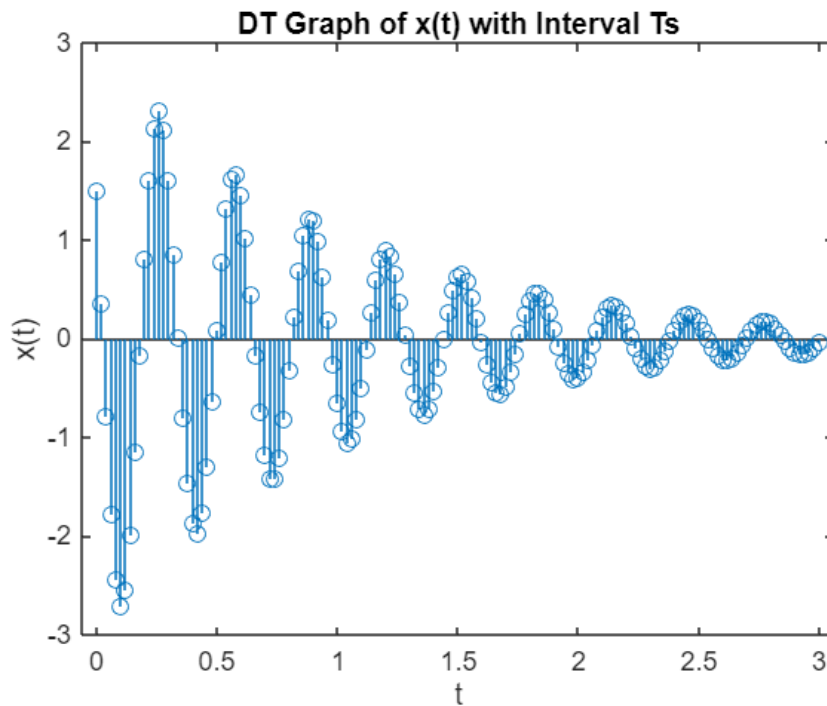


Task 2b

```

Fs = 50; %50hz ;
Ts = 1/Fs;
t = 0:Ts:3;
x = 3* exp(-t) .*cos(20*t + pi/3);
figure;
stem(t, x)
title('DT Graph of x(t) with Interval Ts')
xlabel('t')
ylabel('x(t)')

```

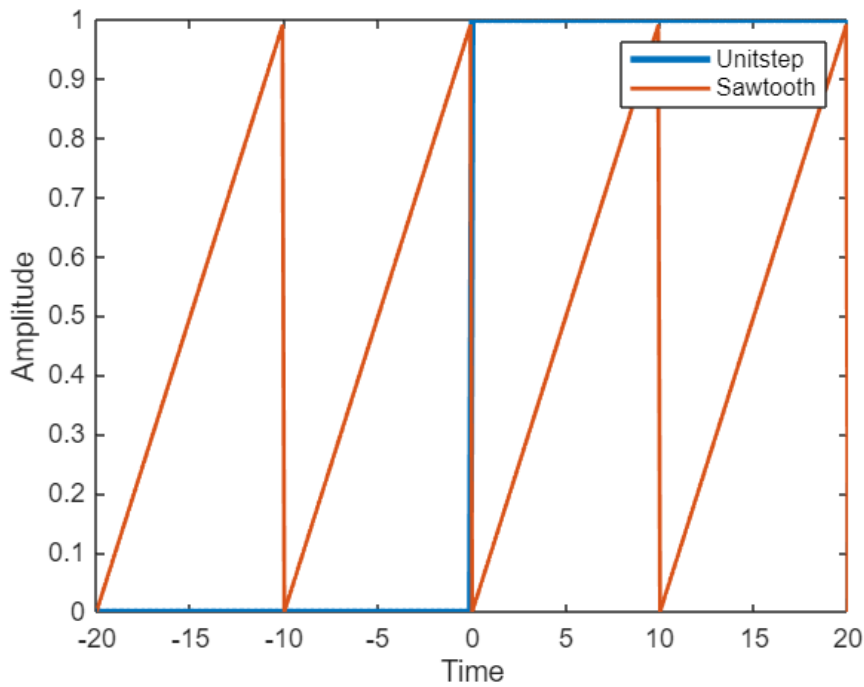


Try 3

```

clc;
clear
t = -20:0.1:20;
unitstep = t >= 0; % getting 1 when t is positive
s = 0.5*(sawtooth(2*pi*0.1*t)+1); % applying transformations to make the graph
start from the origin and have 1 amplitude
figure;
plot(t, unitstep, 'linewidth', 2.5)
xlabel('Time');
ylabel('Amplitude');
hold on; %retains plots in the current axes so that new plots added to the axes do
not delete existing plots
plot(t, s, 'linewidth', 1.5)
xlabel('Time');
ylabel('Amplitude');
xlim([-20 20]) % limits of time
ylim([0 1]) % limits of the amplitude of the graphs
legend('Unitstep', 'Sawtooth')
hold off;

```

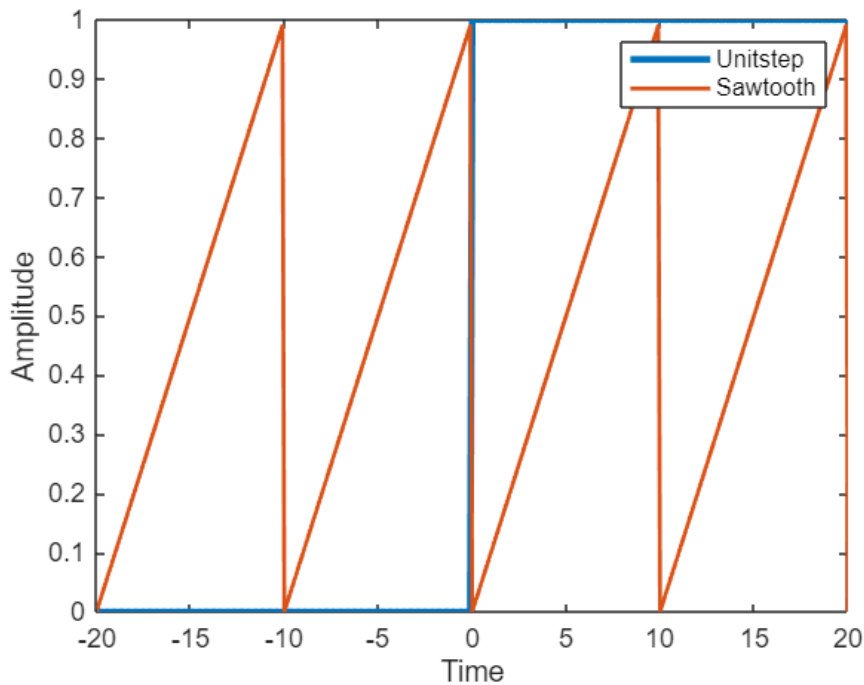


Task 3

```

clc; clear
t = -20:0.1:20;
unitstep = t >= 0; % getting 1 when t is positive
s = 0.5*(sawtooth(2*pi*0.1*t)+1); % applying transformations to make the graph
start from the origin and have 1 amplitude
figure;
plot(t, unitstep, 'linewidth', 2.5)
xlabel('Time');
ylabel('Amplitude');
hold on;
plot(t, s, 'linewidth', 1.5)
xlabel('Time');
ylabel('Amplitude');
xlim([-20 20]) % limits of time
ylim([0 1]) % limits of the amplitude of the graphs
legend('Unitstep', 'Sawtooth')
hold off;

```

Task 4

```
% Time scale: 0 to 5 seconds with interval of 0.005
t = 0:0.005:5;

% Frequencies
f1 = 1; % Frequency of first sinusoid (1 Hz)
f2 = 10; % Frequency of second sinusoid (10 Hz)

% Generate two unit-amplitude sinusoids
xt1 = sin(2*pi*f1*t); % Sinusoid with frequency f1
xt2 = sin(2*pi*f2*t); % Sinusoid with frequency f2

% Generate noise with the same number of elements as xt1/xt2
noise = randn(size(t)); % Random noise

% Combine signals with noise
yt = xt1 + xt2 + 0.2*noise; % Resulting noisy signal

% Filtered signals
N1 = 5; % Number of samples for averaging filter (N=5)
zt1 = filter(ones(1, N1)/N1, 1, yt); % Filtered output for N=5

N2 = 20; % Number of samples for averaging filter (N=20)
zt2 = filter(ones(1, N2)/N2, 1, yt); % Filtered output for N=20

% Plotting
figure;
```

```

% Plot 1: xt1 versus time
subplot(3, 2, 1); % Position (1st row, 1st column)
plot(t, xt1, 'LineWidth', 1.5); % Plot the first signal
title('Plot 1: x_1 vs Time');
xlabel('Time (s)');
ylabel('Amplitude');
xlim([0 5])
ylim([-1 1])

% Plot 2: xt2 versus time
subplot(3, 2, 2); % Position (1st row, 2nd column)
plot(t, xt2, 'LineWidth', 1.5); % Plot the second signal
title('Plot 2: x_2 vs Time');
xlabel('Time (s)');
ylabel('Amplitude');
xlim([0 5])
ylim([-1.25 1.25])

% Plot 3: 0.2*noise versus time
subplot(3, 2, 3); % Position (2nd row, 1st column)
plot(t, 0.2*noise, 'LineWidth', 1.5); % Plot the noise signal
title('Plot 3: 0.2*noise vs Time');
xlabel('Time (s)');
ylabel('Amplitude');
xlim([0 5])
ylim([-0.5 0.5])

% Plot 4: yt versus time
subplot(3, 2, 4); % Position (2nd row, 2nd column)
plot(t, yt, 'LineWidth', 1.5); % Plot the noisy signal
title('Plot 4: y_t vs Time');
xlabel('Time (s)');
ylabel('Amplitude');
xlim([0 5])
ylim([-2 2])

% Plot 5: zt for N=5 versus time
subplot(3, 2, 5); % Position (3rd row, 1st column)
plot(t, zt1, 'LineWidth', 1.5); % Plot the filtered signal (N=5)
title('Plot 5: z_t(N=5) vs Time');
xlabel('Time (s)');
ylabel('Amplitude');
xlim([0 5])
ylim([-2 2])

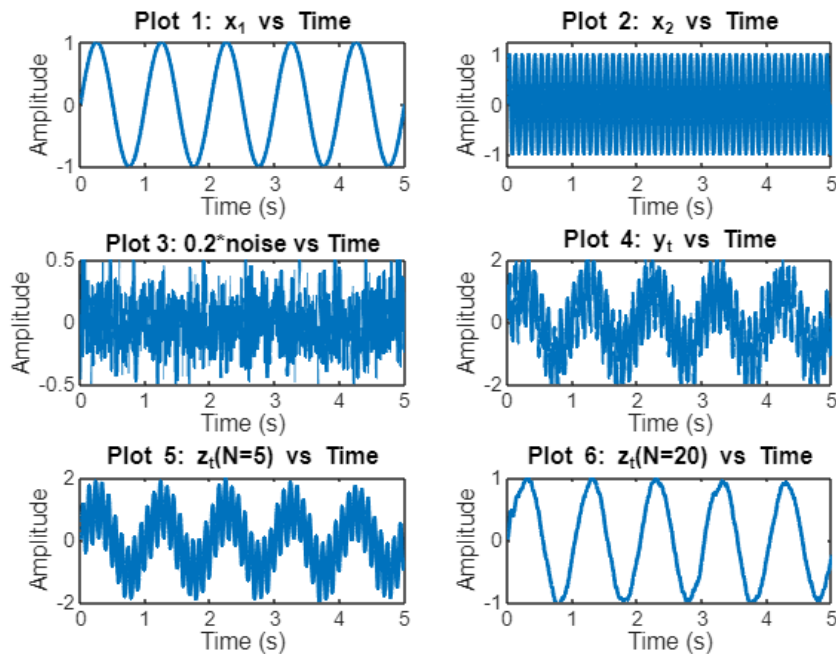
% Plot 6: zt for N=20 versus time
subplot(3, 2, 6); % Position (3rd row, 2nd column)
plot(t, zt2, 'LineWidth', 1.5); % Plot the filtered signal (N=20)
title('Plot 6: z_t(N=20) vs Time');

```

```

xlabel('Time (s)');
ylabel('Amplitude');
xlim([0 5])
ylim([-1 1])

```



Task 5-1a

```

x = 10.09 + 55.3i;
d = deg2rad(85.5); %converting to rad
[a,b] = pol2cart(d,20); %using funct to convert to rad
u = a+ 1i*b; % converting into a complex no using different functions to calc real,
c=x+u

```

```

c =
11.6592 +75.2383i

```

```

r3=real(c) %real value

```

```

r3 =
11.6592

```

```

i3 = imag(c) %imaginary value

```

```

i3 =
75.2383

```

```

c3=conj(c) %complex conjugate

```

```

c3 =
11.6592 -75.2383i

```

```

a3 = abs(c) %magnitude

```

```
a3 =  
76.1364
```

```
p3 = angle(c) %phase angle
```

```
p3 =  
1.4171
```

Task 5-1b

```
x = -4+10i;  
q = deg2rad(10); %converting to rad  
[a,b] = pol2cart(q,15); %using funct to convert to rad  
u = a+ 1i*b; % converting into a complex no  
%using different functions to calc real,  
d= u+x;  
r4=real(d) %real value
```

```
r4 =  
10.7721
```

```
i4 = imag(d) %imaginary value
```

```
i4 =  
12.6047
```

```
c4=conj(d) %complex conjugate
```

```
c4 =  
10.7721 -12.6047i
```

```
a4 = abs(d) %magnitude
```

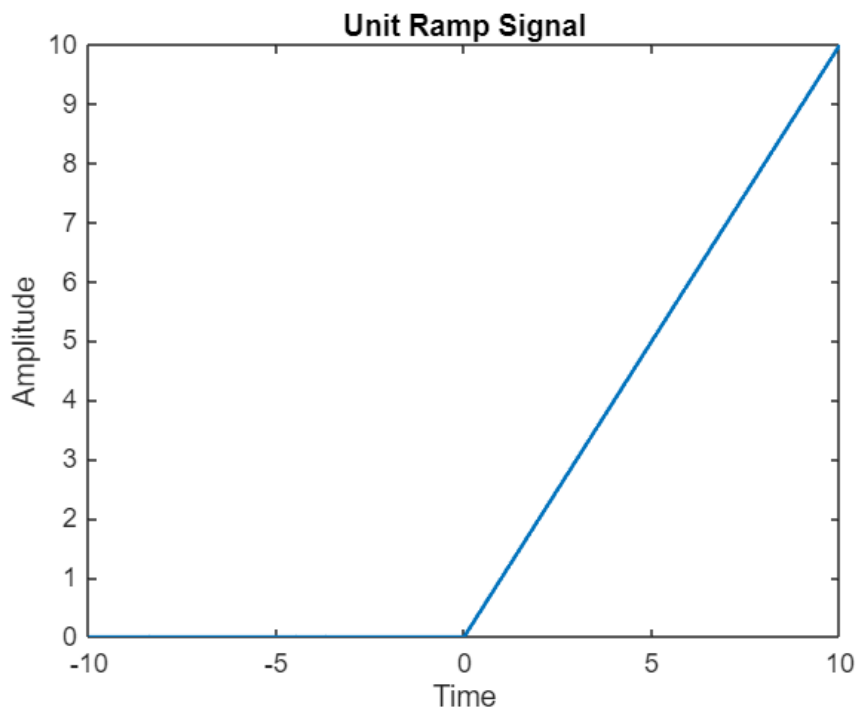
```
a4 =  
16.5806
```

```
p4 = angle(d) %phase angle
```

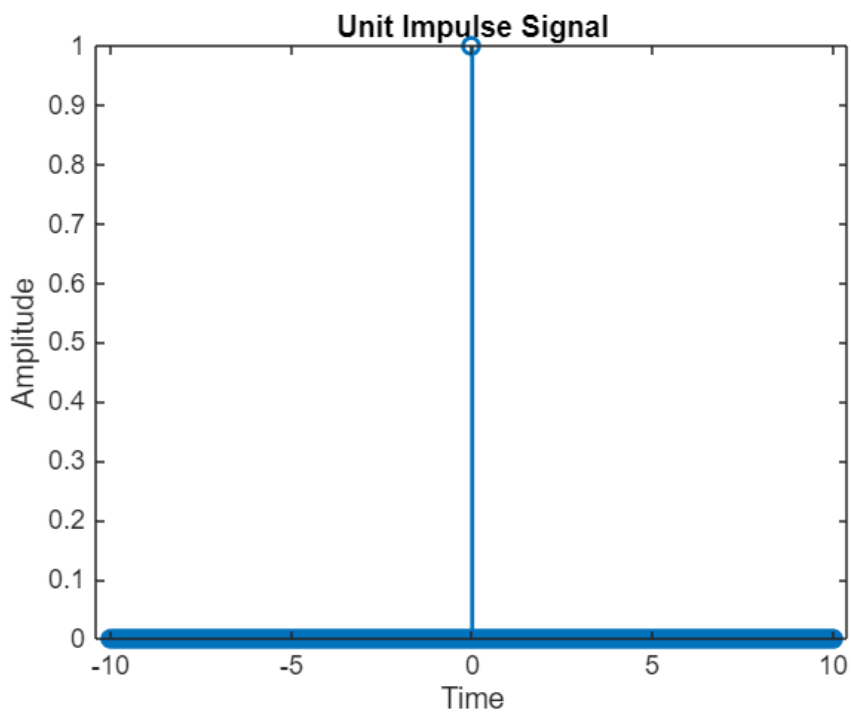
```
p4 =  
0.8636
```

Task 5-2

```
t = -10:0.01:10; % defining time  
%unit ramp signals t*1 when t>0  
r = t .* (t >= 0);  
% Plot the unit ramp signal  
figure;  
plot(t, r, 'LineWidth', 1.5);  
title('Unit Ramp Signal');  
xlabel('Time');  
ylabel('Amplitude');
```



```
t = -10:0.01:10; % defining time
i = (t==0); %is one when time would be 0 else 0
figure;
stem(t, i, 'LineWidth', 1.5);
title('Unit Impulse Signal');
xlabel('Time');
ylabel('Amplitude');
```



Task 6

```
clc;
clear;
t = 0:0.001:1;
fm = 1;
fc = 10;
Am = 1;
Ac = 1;
mt = Am * sin(2*pi*fm*t);
ct = Ac*sin(2*pi*fc*t);
st = (1+ 0.5*mt) .* (ct);
figure;
subplot(3, 1, 1);
plot(t, mt, 'LineWidth', 1.5);
title("Sine Wave f = 1Hz");
xlabel("Time");
ylabel('Amplitude');
subplot(3, 1, 2);
plot(t, ct, 'LineWidth', 1.5);
title("Sine Wave f = 1Hz");
xlabel("Time");
ylabel('Amplitude');
subplot(3, 1, 3);
plot(t, st, 'LineWidth', 1.5);
title("Modulated Signal");
xlabel("Time");
ylabel('Amplitude');
```

